Project Problem and Success Test

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 26

Recall and choose your starting point

> Record your starting explanation for Project Problem and Success Test before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Permissions and Human Review

Without opening the old lesson, explain one key idea from Permissions and Human Review. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Evidence and Sources

Without opening the old lesson, explain one key idea from **Evidence** and Sources. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Image and Voice Pipelines

Without opening the old lesson, explain one key idea from Image and Voice **Pipeline**s. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 26

Practise with fading help

> Complete the supported Project Problem and Success Test trace and record the first step you can explain without help.

Use W2: Write an **acceptance case** for Planning the application. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 26

Individual reasoning check

> Check the proposed Project Problem and Success Test explanation against the supplied facts before discussing it.

Use the worked case for Planning the application. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 26

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Plan independently

Plan a 15-minute fictional club chooser. Specify schema, score, duration limit, tie rule, invalid-data behavior, **output** and three **acceptance case**s.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 26

Your learning target

> Connect the Project Problem and Success Test target to a trace and a justified conclusion.

Specify three **acceptance test**s for a recommender: a normal match, a tied score and no qualifying book. Include expected behaviour before implementation.

WORKBOOK · 6 OF 26

Solve the given case

> Solve the supplied Project Problem and Success Test case and show the deciding evidence.

A project recommends from ten approved books using genre and length.

Define **acceptance test**s.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 26

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Define a testable goal for a book recommender serving readers who want short space books. Include a concrete success condition.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 32 Test and reflect

WORKBOOK · 8 OF 26

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Project Problem and Success Test adds to Transparent Book Recommender and keep the evidence.

Freeze the brief before building so later choices can be traced to **requirement**s.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 26

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 26

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Specify three **acceptance test**s for a recommender: a normal match, a tied score and no qualifying book. Include expected behaviour before implementation.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 26

W1 Name the user need

> Name the user need. Apply the supplied Project Problem and Success Test facts and keep your reasoning.

State the purpose and decision owner of the worked chooser.

My reasoning and evidence

WORKBOOK · 13 OF 26

W2 Write an acceptance case

> Write an acceptance case. Apply the supplied Project Problem and Success Test facts and keep your reasoning.

Give a tie case for A and B with score 8 each and valid durations.

My reasoning and evidence

WORKBOOK · 14 OF 26

W3 Repair vague scope

> Repair vague scope. Apply the supplied Project Problem and Success Test facts and keep your reasoning.

Replace the **requirement** always choose the best activity with a checkable one.

My reasoning and evidence

WORKBOOK · 15 OF 26

W4 Plan independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Plan independently. Apply the supplied Project Problem and Success Test facts and keep your reasoning.

Plan a 15-minute fictional club chooser. Specify schema, score, duration limit, tie rule, invalid-data behavior, **output** and three **acceptance case**s.

My reasoning and evidence

WORKBOOK · 16 OF 26

W5 State non-goals

> State non-goals. Apply the supplied Project Problem and Success Test facts and keep your reasoning.

Name two capabilities excluded from this **prototype** and explain why.

My reasoning and evidence

WORKBOOK · 17 OF 26

A reproducible data investigation — W1 State the question

> Use the supplied Project Problem and Success Test task to record your reasoning, evidence and next step.

Write the worked **investigation** question so its population of records is clear.

My reasoning and evidence

WORKBOOK · 18 OF 26

A reproducible data investigation — W2 Trace exclusions

> Use the supplied Project Problem and Success Test task to record your reasoning, evidence and next step.

Classify 0, None, -2 and 60 under the schema.

My reasoning and evidence

WORKBOOK · 19 OF 26

A reproducible data investigation — W3 Repair a denominator

> Use the supplied Project Problem and Success Test task to record your reasoning, evidence and next step.

A report divides a valid total of 24 by six supplied rows, though only three are valid. Correct the valid-record mean.

My reasoning and evidence

WORKBOOK · 20 OF 26

A reproducible data investigation — W4 Investigate independently

> Use the supplied Project Problem and Success Test task to record your reasoning, evidence and next step.

Analyze raw durations [6, 12, None, 0, 70, 18] with the same schema. Report question, accepted list, rejections, mean and one **limitation**.

My reasoning and evidence

WORKBOOK · 21 OF 26

A reproducible data investigation — W5 Support reproduction

> Use the supplied Project Problem and Success Test task to record your reasoning, evidence and next step.

A report gives only mean 9. What additional **evidence** lets a peer reproduce it?

My reasoning and evidence

WORKBOOK · 22 OF 26

Feedback, return check and learning record

Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 23 OF 26

Pictorial case practice: Project Problem and Success Test

> A fictional reading-break chooser needs a transparent contract. Explain the labelled step before checking the changed condition.

A fictional reading-break chooser needs a transparent contract.

Items have id text, minutes integer 0–30, interest integer 0–5. Request limit is integer 0–30.

Eligible if minutes<=limit; score=interest; highest wins; ties choose alphabetically smaller ID; invalid rows excluded and counted.

AI ACADEMY / AI-2/32

Project Problem and Success Test

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Specify inputs

Validate each row and request; keep rejected-row reasons.



2. Filter and rank

At limit 8: A(time8,interest3), B(9,5), C(6,3). B is ineligible.



3. Expected output

A beats tied C by ID; report A and its score 3.



4. Acceptance cases

Also plan invalid request, no eligible item and equal-score cases.

CHANGE ONE THING

Reduce the limit from 8 to 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Write an equality test and explain which **requirement** it checks.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Retain the Transparent Book Recommender's schema, constraints, score, tie rule and acceptance table as its build specification.

WORKBOOK · 24 OF 26

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Retain the Transparent Book Recommender's schema, constraints, score, tie rule and acceptance table as its build specification.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 25 OF 26

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional reading-break chooser needs a transparent contract.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Also plan invalid request, no eligible item and equal-score cases.

1. Filter and rank

2. Specify **input**s

3. **Acceptance case**s

4. Expected **output**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Acceptance case**s Also plan invalid request, no eligible item and equal-score cases. Planning fixes what success means before code returns an answer.

The duration boundary is including equality, and the tie policy prevents accidental list-order dependence. A complete **output** may include why other rows were excluded.

Test expectations should come from the contract rather than copying the first program result.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Reduce the limit from 8 to 5. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Return no eligible item.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Return no eligible item. All

three valid rows take more than five minutes.

Repair a missing building block

Review Permissions and Human Review (ai-2/31). Goal: Explain the **mechanism** and terminology of least privilege and tool **scope**s. Short practice: Which proposal is allowed, and why is an authorised object alone insufficient?.

Return to this week and explain what you changed.

Review **Evidence** and Sources (ai-2/29). Goal: Explain the **mechanism** and terminology of claim support and citation evaluation.

Short practice: Why is the original answer not fully supported despite having citations on every claim?. Return to this week and explain what you changed.

Review Image and Voice **Pipeline**s (ai-2/26). Goal: Explain the **mechanism** and terminology of speech **pipeline**s and multimodal **evidence**.

Short practice: Which stage first fails, and why should the faulty transcript not be silently overwritten?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

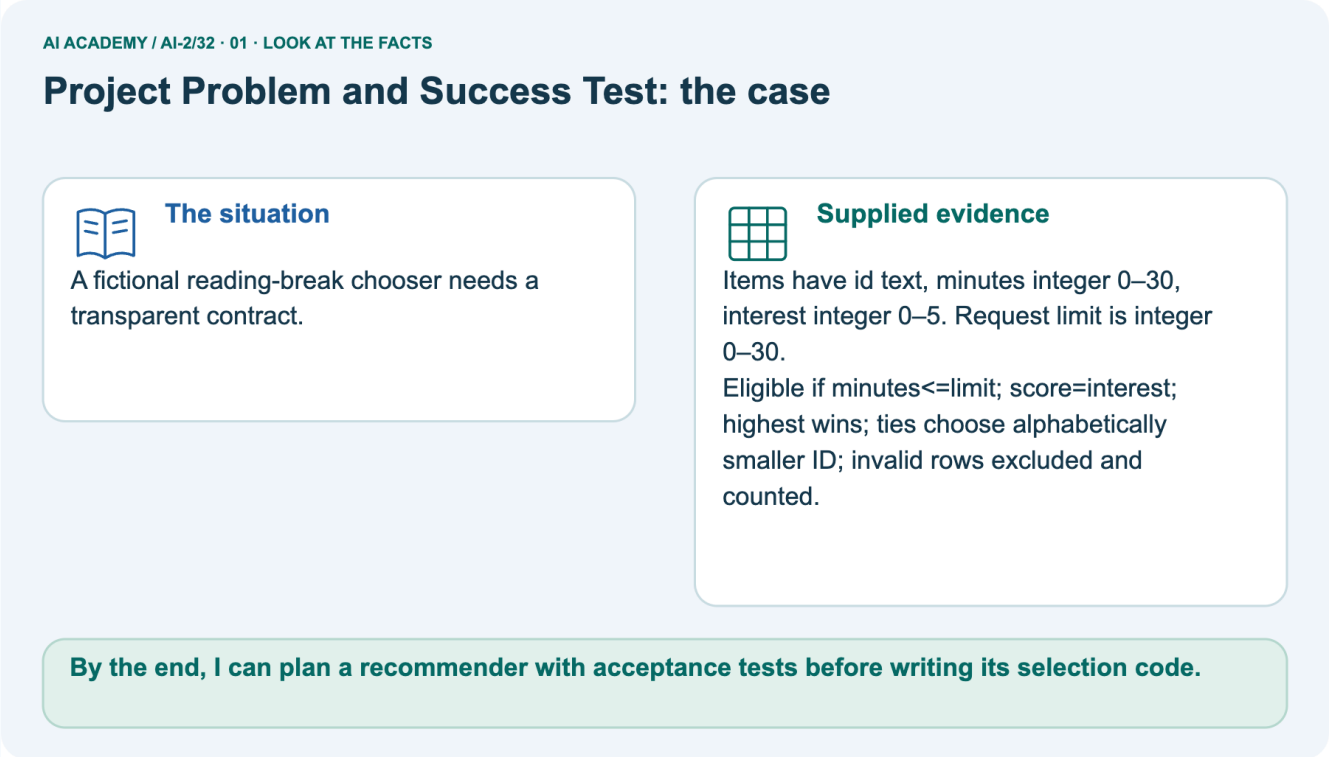
What I repaired, the evidence I kept and the next test

WORKBOOK · 26 OF 26

Visual concept journey

> Explain each Project Problem and Success Test picture from its labels, then record a relevant limit.

01. Project Problem and Success Test: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

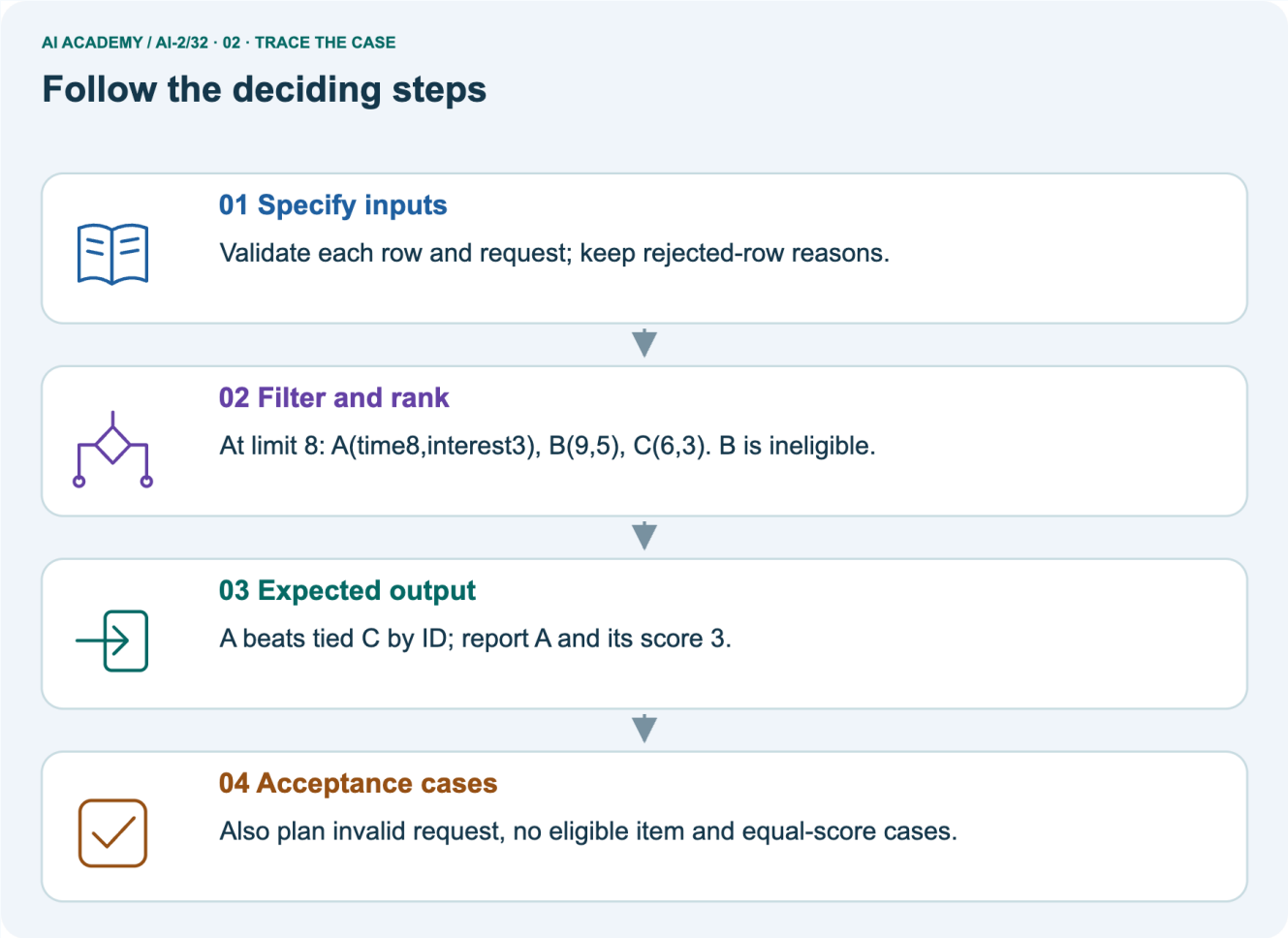
A fictional reading-break chooser needs a transparent contract. Supplied facts: Items have id text, minutes integer 0–30, interest integer 0–5. Request limit is integer 0–30.

Eligible if minutes<=limit; score=interest; highest wins; ties choose alphabetically smaller ID; invalid rows excluded and counted.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

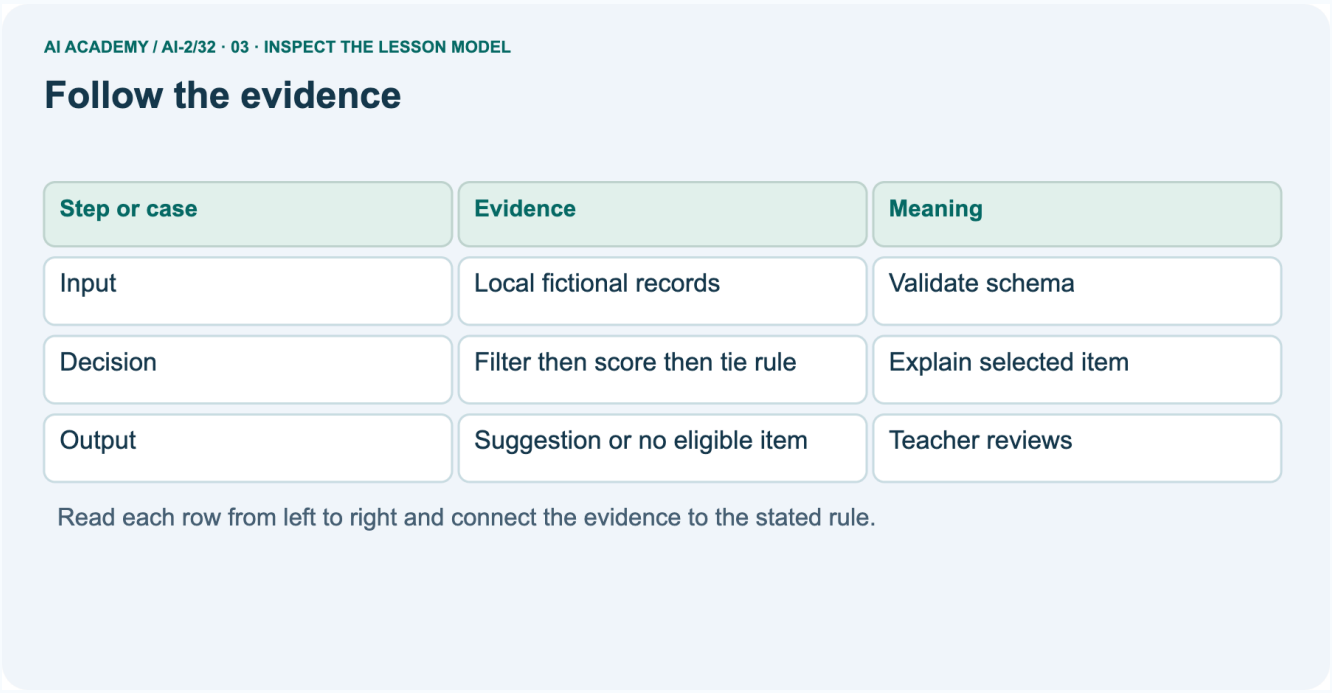
Specify **input**s: Validate each row and request; keep rejected-row reasons. → Filter and rank: At limit 8: A(time8,interest3), B(9,5), C(6,3).

B is ineligible. → Expected **output**: A beats tied C by ID; report A and its score 3. → **Acceptance case**s: Also plan invalid request, no eligible item and equal-score cases.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow the evidence



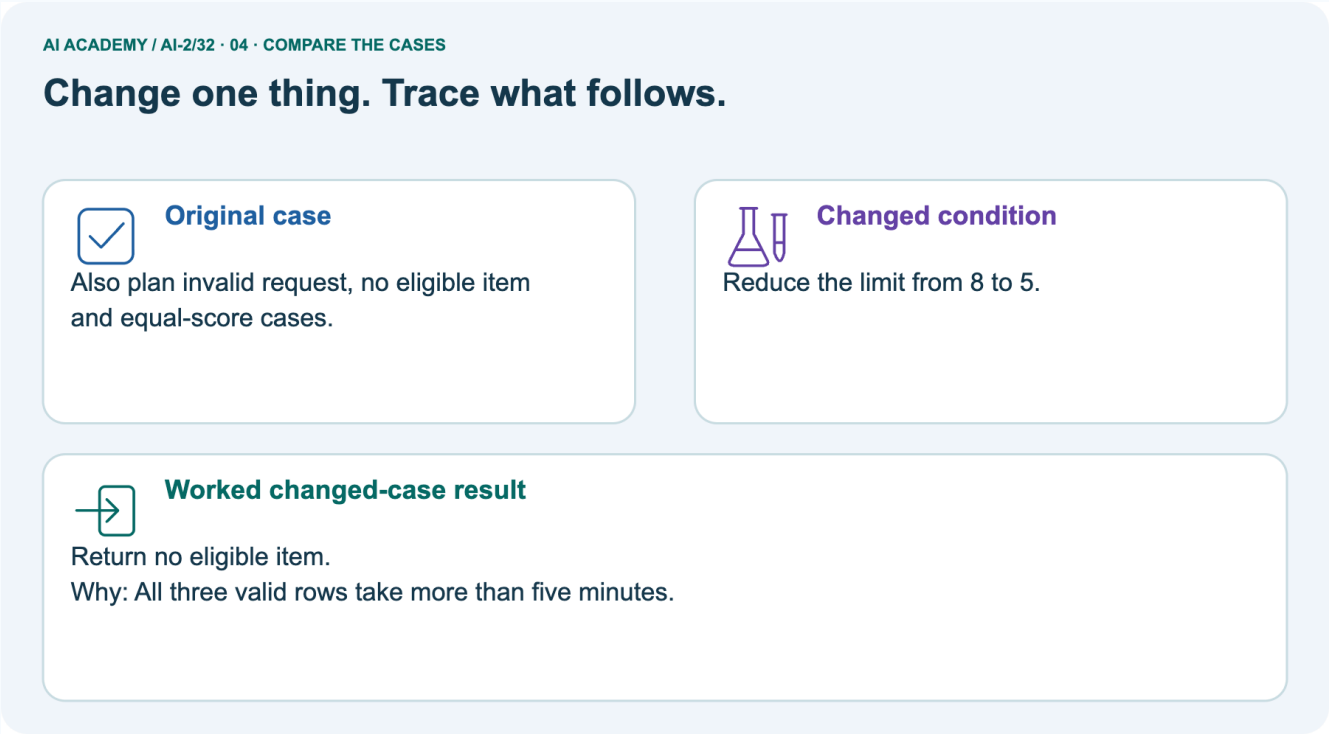
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. **Input** | Local fictional records | Validate schema. Decision | Filter then score then tie rule | Explain selected item. **Output** | Suggestion or no eligible item | Teacher reviews

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

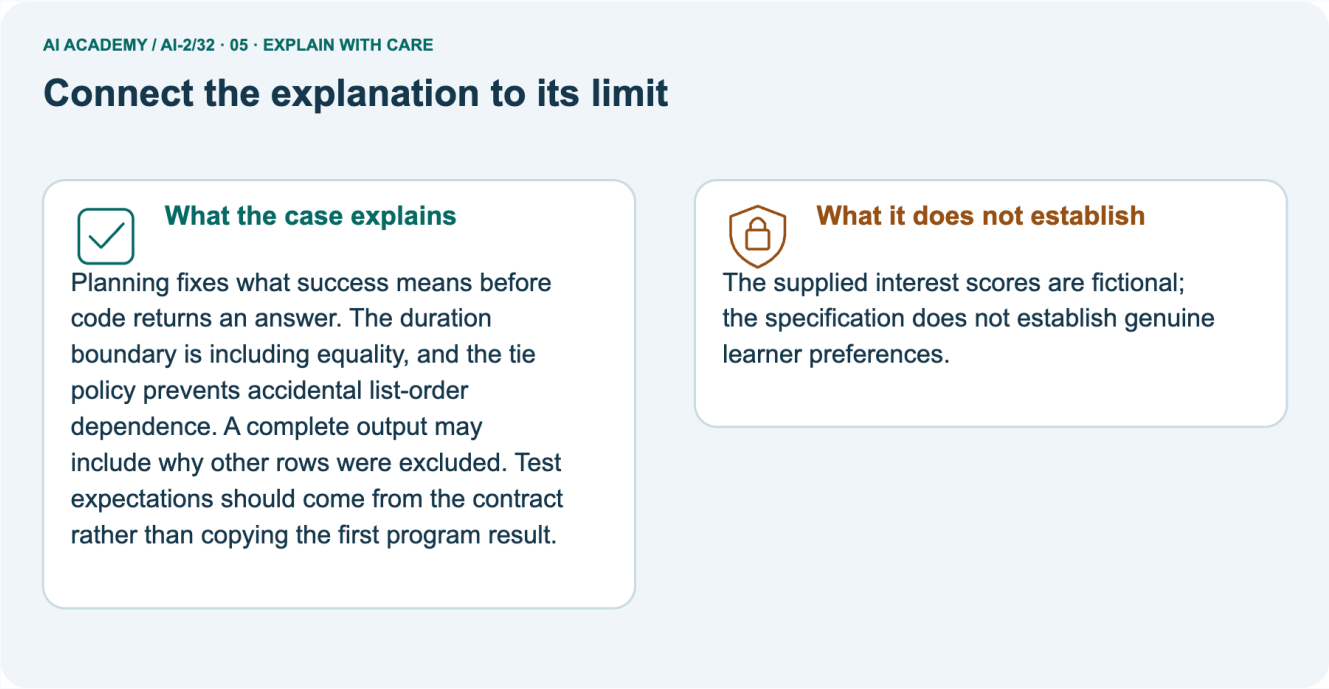
Original case: Also plan invalid request, no eligible item and equal-score cases. Changed condition: Reduce the limit from 8 to 5. Expected result: Return no eligible item.

Reason: All three valid rows take more than five minutes.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Planning fixes what success means before code returns an answer. The duration boundary is including equality, and the tie policy prevents accidental list-order dependence.

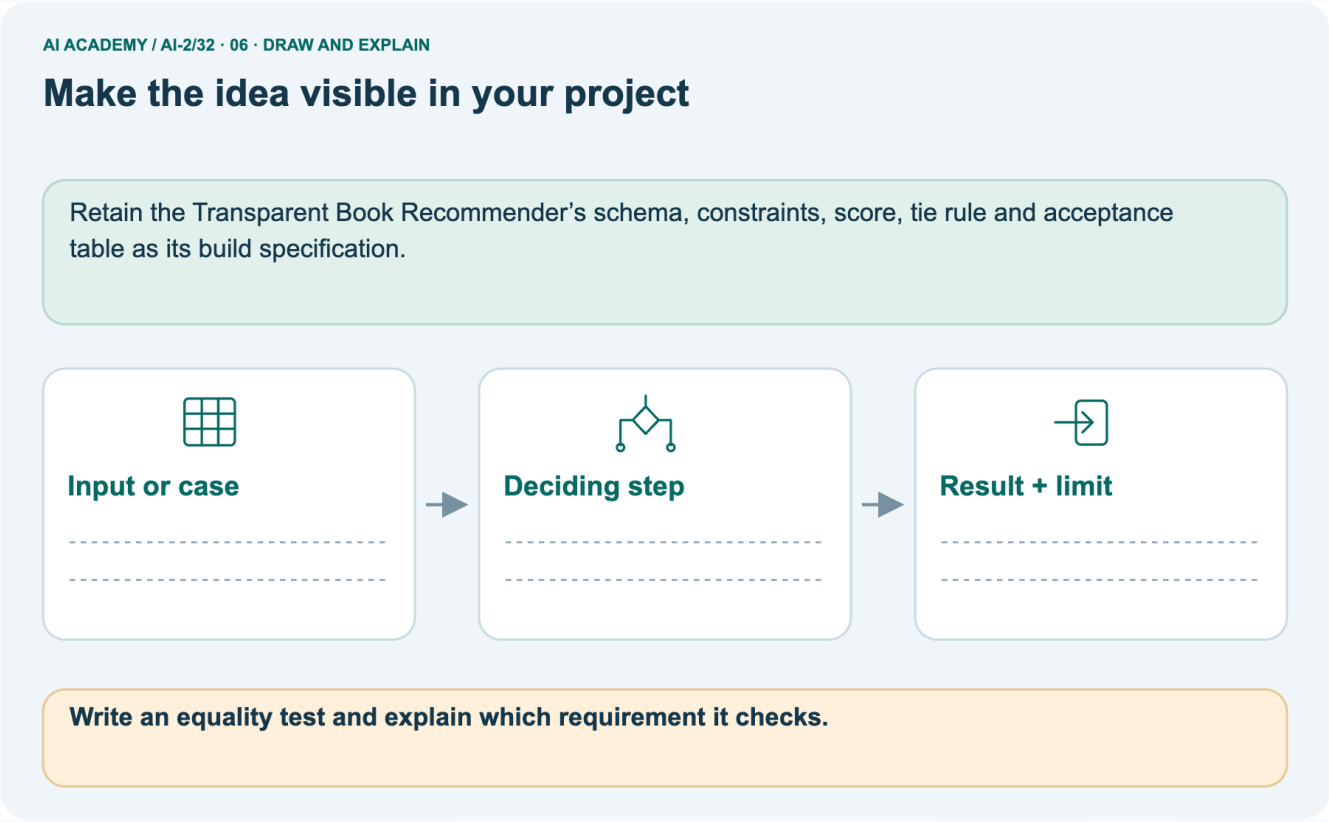
A complete **output** may include why other rows were excluded. Test expectations should come from the contract rather than copying the first program result.

Limit: The supplied interest scores are fictional; the specification does not establish genuine learner preferences.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Retain the Transparent Book Recommender's schema, constraints, score, tie rule and acceptance table as its build specification. Draw your **input**, deciding step, result and limit.

Supported question: Write an equality test and explain which **requirement** it checks.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.